

1. Introduction.....	2
2. Relative Clause in Malayalam.....	2
3. LFG Analysis	5
4. The Implementation of Relative Clause.....	6
5. Analysis of Example Relative Clause.....	7
6. Limitation.....	9
7. Conclusion.....	9
Reference.....	10
Appendix.....	11

1. Introduction

This is a documentation of implementing Malayalam relative clauses using the Lexical-Functional Grammar (LFG) framework. This grammar tries to describe how relative clauses combine with nouns and how other grammatical functions of a missing element inside a relative clause can be represented.

Malayalam, which is one of the four major Dravidian languages spoken in the southern part of India, Kerala, is a head-final, agglutinative SOV language. It also allows scrambling. The language lacks an overt subject-verb agreement.

Malayalam relative clauses have a prenominal or N-final relative clause, like its sister Dravidian languages (Hima, 2017). Unlike English, which uses relative pronouns such as "who," "which," and "that" for creating relative clauses, Malayalam uses the participial form of the verb rather than overt relative pronouns.

For instance, the phrase “vanna kutti” is understood as “the child who came.” Here, “vanna” is the participle form of the verb meaning "came," and “kutti” means “child”. The person who came and the child refer to the same entity here. Therefore, relative clause contains a gap corresponding to the subject of the verb.

The grammar developed establishes two important configurations—subject-gap and object-gap relative clauses. Separate alternatives are used in the implementation to represent the relationship between the relative clause and the modified noun through XLE.

The grammar also consists of other components, including noun, pronoun, demonstrative, numeral, adjective, adverb, and prepositional phrase verb rules, which provides an environment for relative clauses to occur. However, the implementation of the relative clause was the main focus of this report.

2. Relative Clause in Malayalam

2.1 General Properties

Relative clauses, which are part of sentences, describe a noun. The referent of the noun will be either restricted or described by them. A relationship between a noun and an argument is established inside the clause by a relative clause.

In Malayalam, the relative clause is marked in the verb by a morpheme, "-a," which is classified in De Vries's (2002) typology of relative clauses as a relative marker, as opposed to a relative pronoun or complementizer.

2.2 The relativizer “-a”

Hima (2017) describes “-a” as the relative marker or relativizer in the literature on Malayalam relative clauses. She identifies two important characteristic features of “-a”:

- (i) It occurs in the verb in the relative clauses, participles, and adjectives, but not in the sentential complement structure.
- (ii) It cannot stand alone without a nominal following it.” (p. 136).

Consider the examples given below:

- (1) Vann-a kutti
 come-RELCL child
 The child who came.
- (2) Ammu vayich-a pusthakam
 Ammu read-RELCL book
 The book that Ammu read.

In example (1), the subject of “come” is interpreted as “Kutti” (child). There is no overt noun corresponding to the subject in the relative clause. Here, the participle verb “vanna” supplies the verbal meaning, while the noun (“kutti”) that is being modified fills the semantic subject role.

In example (2), the subject is overtly expressed as the subject of the relative clause, which is “Ammu”. But the object of “vayicha” (read) is not overtly expressed. The semantic subject role, which is missing, is being fulfilled by “pusthakam” (book).

2.3 Subject-gap relative clauses

This is the type of clause where the modified noun acts as the subject of the relative clause itself. As the noun is doing the action, the grammatical subject is missing or gapped from its usual position.

Example (1) can be interpreted as “the child who came.”

The lexical item *vanna* is identified in the grammar as;

vanna VREL * @ (INTRANS come)
 @PART.

The template INTRANS gives the verb the predicate.

(^ PRED) = 'come<(^SUBJ)>'

While @PART specifies that the verb is participial:

(^ VFORM) = part

The modified noun is introduced by the NP rule:

RELCL: (! SUBJ)=^;

...

N: ^=!

The important equation is:

RELCL: (! SUBJ)=^

This here states that the f-structure of the noun phrase provides the subject for the relative clause. In other words, the noun that is outside of the relative clause fills the subject function of the relative clause.

The resulting analysis would be that the noun “kutti” will simultaneously be the head noun of the NP as well as the semantic subject associated with the predicate “come.”

2.3 Object-gap Relative clause

Consider the below example:

Ammu vayicha pusthakam
Ammu read-RELCL book
The book that ammu read.

Here, the overt subject of the relative clause is Ammu. The verb “vayicha” (read) is transitive, but the object is not present overtly inside the relative clause as it corresponds to the noun “pusthakam”.

The relative clause rule contains the relevant subject and object positions:

RELCL --> ...
 (NP: (^ SUBJ)=!
 (! CASE)=nom)
 (NP: (^OBJ)=!
 (!CASE)=acc)
 ...
 VREL: ^=!.

The connection with the noun that is modified is made in the NP rule:

RELCL: (! OBJ)=^;

This means that the f-structure of the NP is identified with the object of the relative clause.

So, the object of “read” is filled by "pusthakam", even though it does not occur inside the relative clause as an overt object.

3. LFG Analysis

3.1 The NP rule

```
NP --> {  
    AP*: ! $ (^ ADJUNCT);  
    NMOD*: ! $ (^ ADJUNCT);  
    N: ^=!  
  
    |  
  
    DEM*  
    NUMP*  
    AP*: ! $ (^ ADJUNCT);  
    NMOD*: ! $ (^ ADJUNCT);  
    N: ^=!  
  
    |  
    RELCL: (! SUBJ)=^  
            ! $ (^ ADJUNCT);  
    DEM*  
    NUMP*  
    AP*: ! $ (^ ADJUNCT);  
    NMOD*: ! $ (^ ADJUNCT);  
    N: ^=!  
  
    |  
    RELCL: (! OBJ)=^  
            ! $ (^ ADJUNCT);  
    DEM*  
    NUMP*  
    AP*: ! $ (^ ADJUNCT);  
    NMOD*: ! $ (^ ADJUNCT);  
    N: ^=!  
    PRON: ^=!  
    }.
```

"attributive nominal modifiers"

"optional demonstrative"

"optional numeral"

"subject-gap relative clause"

"object-gap relative clause"

The first alternative allows ordinary noun phrases, while the 3rd and 4th specifically allows relative clauses.

The subject-gap option is:

RELCL: (! SUBJ)=^;

The object-gap option is:

RELCL: (! OBJ)=^;

These two rules create the core of the relative-clause implementation.

3.3 Relative Clause as an NP modifier

Here, relative clauses are not treated as independent sentences; instead, they are embedded inside the NP.

The general structure of the NP is:

[RELCL] [DEM] [NUMP] [AP] [NMOD] [N]

The connection of the relative clause with the noun is done through the functional equation in the NP rule.

This important structure captures the semantic relationship between the noun and the relative clause with the requirement of a grammar to represent the overt relative pronoun.

4. The Implementation of Relative Clause

The RELCL rule implemented as:

```
RELCL --> (PP: ! $ (^ ADJUNCT))
          (ADV: ! $ (^ ADJUNCT))
          (NP: (^ SUBJ)=!
            (! CASE)=nom)

          ( { (NP:(^ OBL) =!
              (!CASE) = dat)
            |
              (NP:(^ OBJ2) =!
                (!CASE) = dat) } )

          (ADV: ! $ (^ ADJUNCT))
          VREL: ^=!.
```

As seen in the rule, there are several components inside the relative clause.

First there is an optional preposition phrase followed by an optional adverb, which can appear before the subject.

Then the rule says that a relative clause can contain an overt nominative subject or an overt accusative object, or it allows for a dative argument. Finally, the relative clause ends with a relative participial verb.

This is important for the Malayalam data represented in the grammar because relative clauses use participial verb forms such as *vanna*, *vayicha*, *vangiya*, and *ezhuthiya*.

5. Analysis of Example Relative Clause

5.1 Example from Subject-gap:

Consider:

vanna kutti

The intended interpretation is:

“the child who came”

The lexical entries are:

vanna VREL * @(INTRANS come)
@PART.

kutti N * @(NOM-CN child sg).

The noun phrase is analyzed using the subject-gap alternative:

NP --> RELCL: (! SUBJ)=^;
...
N: ^=!

The relative clause contains:

VREL: ^=!.

The lexical entry for “*vanna*” supplies:

PRED = 'come<(^SUBJ)>'
VFORM = part

The equation:

(! SUBJ)=^

identifies the subject of the relative clause with the f-structure of the containing NP.

The noun “kutti” does not need to be represented as an overt subject inside the RELCL. Its f-structure is identified with the subject function of the relative predicate.

5.2 Example from Object-gap:

Now consider:

Ammu vayicha pusthakam

The intended interpretation is:

“the book that Ammu read”

The relevant lexical entries are:

Ammu N * @(NAME ammu fem).

vayicha VREL * @(TRANS read.

@PART.

pusthakam N * @(NOM-CN book sg).

The relative clause contains the overt subject *Ammu*. The verb *vayicha* is transitive and therefore requires both a subject and an object.

The relative clause itself provides:

NP: (^ SUBJ)=!

for *Ammu*, and the predicate is:

read<SUBJ, OBJ>

The object is not overtly realized inside the relative clause. Instead, the NP rule uses:

RELCL: (! OBJ)=^;

This identifies the object of the relative clause with the noun phrase being constructed.

6. Limitation

One major limitation is that the current relative clause rules are good for only a specific constituent order. Malayalam has fairly free word order. A more comprehensive Malayalam grammar would need to account for the flexibility in the word order and the additional types of relative clause.

Additionally, the grammar does not provide a complete morphological analysis of Malayalam relative clauses, as it requires a morphological analyzer to identify and analyze each morpheme. In the current grammar, lexical entries for words associated with case, number, and tense are entered separately in the lexicon.

7. Conclusion

The grammar demonstrates how LFG can model Malayalam relative clauses by separating constituent structure from functional structure. Rather than representing the relation through movement, the grammar uses functional equations to identify the grammatical function inside the relative clause with the f-structure of the modified noun. This provides a compact and linguistically motivated analysis of the relative-clause phenomenon and provides a foundation for extending the grammar to a larger range of Malayalam constructions.

Reference

1. Hima, S. (2017). Relative clauses and/or participials in Malayalam. *Language in India*, 17(7).
2. Asher, R. E., & Kumari, T. C. (1997). *Malayalam*. Routledge.

Appendix

Abbreviations Used

Abbreviation Meaning

ACC	accusative
COP	copula
DAT	dative
DEM	demonstrative
GEN	genitive
PL	plural
PST	past
REL	relative clause/relative participle

General Pattern

The Malayalam relative-clause construction illustrated here follows the pattern:

[Relative Clause] + [Head Noun]

For example:

aval ezhuthi-ya kathu
she write-REL letter

The relative clause *aval ezhuthiya* precedes its head noun *kathu* (“letter”).

Sentences That Should Work

1. *aval ezhuthiya kathu vannu*

aval ezhuthi-ya kathu vann-u
she write-REL letter come-PST
The letter that she wrote came

2. *aval innale ezhuthiya kathu vannu*

aval innale ezhuthi-ya kathu vann-u
she yesterday write-REL letter come-PST
The letter that she wrote yesterday came.

3. innale aval ezhuthiya kathu vannu

innale aval ezhuthi-ya kathu vann-u
yesterday she write-REL letter come-PST
The letter that she wrote yesterday came.

4. innale kuracha naaya

innale kurach-a naaya
yesterday bark-REL dog
The dog that barked yesterday

5. paadiya kutti chirichu

paadi-ya kutti chirich-u
sing-REL child laugh-PST
The child that sang laughed

6. paadiya kuttikal chirichu

paadi-ya kutti-kal chirich-u
sing-REL child-PL laugh-PST
Children who sang laughed.

7. njan vangiya pusthakam nallathan

njan vaangi-ya pusthakam nallath-aan
I buy-REL book good-COP
The book that I bought is good.

8. adutha pareekshaku varunna chodhyangal

adutha pareeksha-kku varunn-a chodhyam-gal
next exam-DAT come-REL question-PL
The question that will come for next exam.

9. innale vanna aal

innale vann-a aal

yesterday come-REL person

The person who came yesterday.

10. Swathi Punnyakk kodutha pusthakam

Swathi Punnya-kku koduth-a pusthakam

Swathi Punnya-DAT give-REL book

The book that Swathi gave to Punnya.

11. enikk kittiya aa rand velli paathrangal

enikk kitt-iyaa aa rand velli paathram-kal

I.DAT get-REL DEM two silver vessel-PL

The silver vessel that I got.

12. patti poochaye kadichu

patti poocha-ye kadich-u

dog cat-ACC bite-PST

The dog bite the cat

13. chaakine mel kedakkunna poocha

chaak-inte mel kidakkunn-a poocha

sack-GEN on lie-REL cat

The cat that lies on the sack.

Sentences That Should Not Work

1. Missing Head Noun

njan ezhuthi-ya

I write-REL

Problem: The relative clause has no overt head noun.

2. Relative Participle After the Noun

pusthakam njan vaangi-ya
book I buy-REL

Problem: The relative participle/relative clause occurs after the head noun. The relative clause should precede the head noun.

3. Relative Clause with No Noun

kand-a
see-REL

Problem: There is no overt head noun.

4. Incorrect Ordering of Adjective and Oblique NP

pareeksha-kku adutha varunn-a chodhyam-kal
exam-DAT next come-REL question-PL

Problem: The ordering is incorrect. The adjective should precede the oblique NP:

5. Attributive Noun Must Precede the Head Noun

aa rand paathram-kal velli
DEM two vessel-PL silver

Problem: The attributive noun/adjective velli occurs after the head noun.

6. Object Without Case Marking

patti poocha kadich-u
dog cat bite-PST

Problem: The object poocha lacks accusative marking.

patti poocha-ye kadich-u
dog cat-ACC bite-PST

